Traffic analysis with Vissim: Computer exercise for the Simulation module of the Infra informatics PhD course

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Traffic analysis with Vissim

1 Purpose

This computer exercise aims to give you some insights into how we can model traffic networks using a commercial microscopic simulation tool called Vissim. Vissim was developed by the German company PTV and is used worldwide to model traffic on a microscopic level. It is a microscopic, time-discrete, and behavior-based simulation model for realistic modeling of city and interurban traffic. During this lab session, you will become familiar with Vissim and learn the basics of creating and running a microscopic simulation model.

Your task is to create and run a simulation model of a traffic intersection in an urban area. More specifically, you will model the intersection between Kungsgatan and Slottsgatan located in the northern part of Norrköping city center (see Figure 1). In Figure 2, the overall intersection layout and the input data of vehicle flow in one hour are illustrated. Material for this exercise is provided as zip-file. Copy the whole directory to your home directory before the first use (you will need all files in the folder). Note that the model implemented in this exercise is a simplification of reality in order to serve pedagogic purposes.

2 Starting Vissim

For more specific information about Vissim, you can access the Vissim manual on Lisam, under the *Help* function in the program, or in the following program directory: *PTV Vision\PTV Vissim 2026\Doc\Eng*. For further information regarding Vissim and PTV Group AG you can visit `https://www.ptvgroup.com/en/solutions/products/ptv-vissim/`.

In this document, the menu, tab choices, object types, and dialog boxes are marked using *italic* font, e.g., save the network (*File - Save as*). In general, you should exit all dialog boxes using *OK* to make sure that your settings are applied to the model.

Start Vissim from the Windows start menu. A network license should be linked automatically. Then, open the *Vissim_Lab.inpx* file. If you open Vissim for the first time, you will be asked to define the right-click behavior. If you would like to check or change the right click behaviour this is done at the *Edit menu, User Preferences 2* and in the in the left window section, select *User interface, Network editor*. Select the option *Right click opens the context menu, New objects are created with CTRL+right click*.



Figure 1: Location of the modeled section.

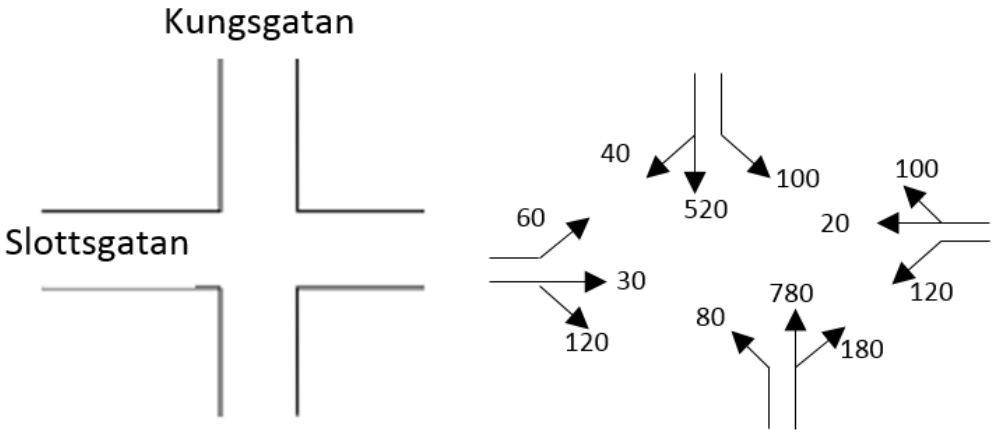


Figure 2: Intersection layout and vehicle inputs.

3 Graphical User Interface (GUI)

The main windows of the Vissim GUI are the *network editor*, the *network objects* sidebar, the *quick view* window, and the *list* window (see Figure 3).

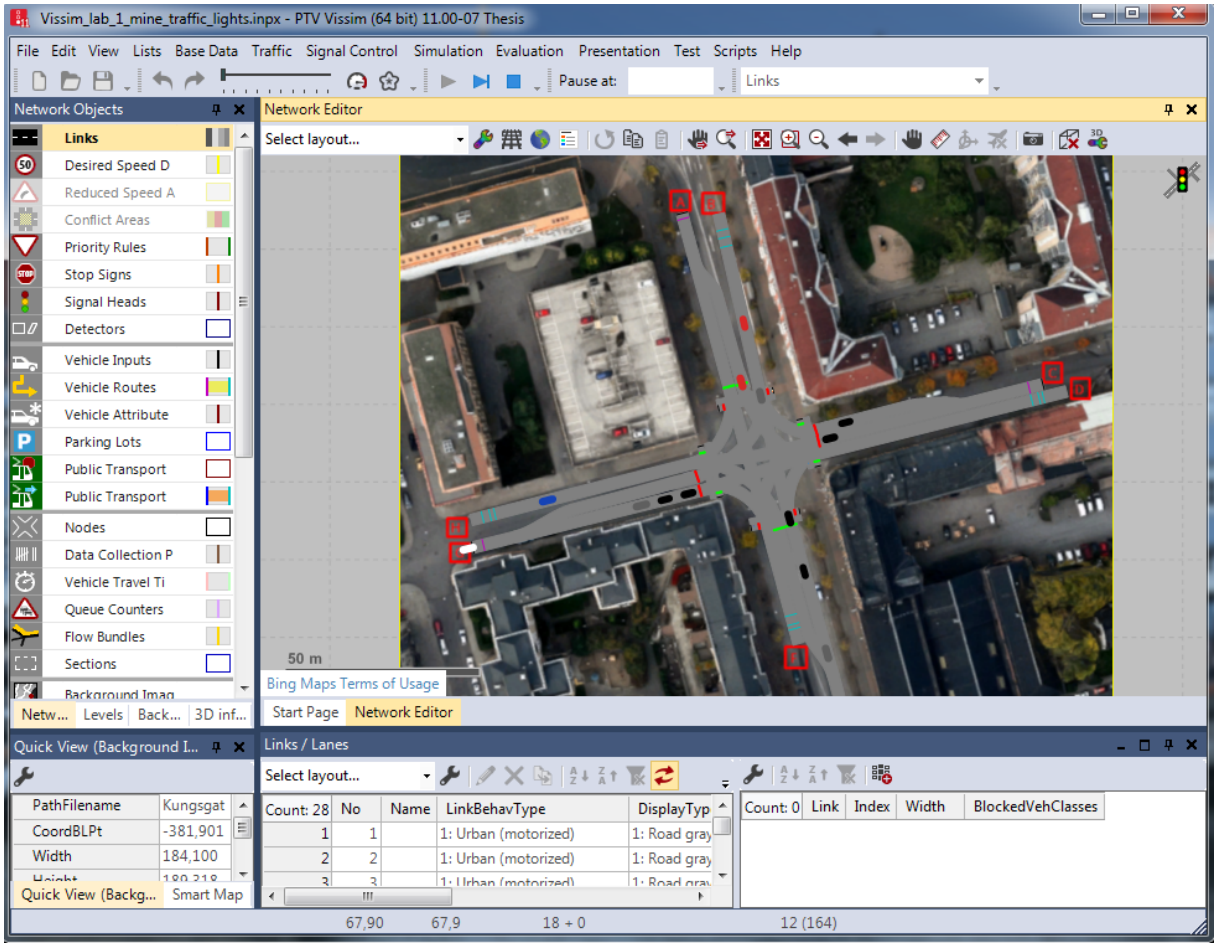
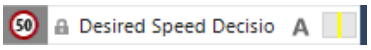


Figure 3: The Graphical User Interface.

The *network editor* displays and defines the network you are working with. The *network objects* sidebar contains a list of all types of network objects you can add to the model. Each object has five settings:



- the *visibility mode*, which hides/shows objects of this type,
- the *selectability mode*, which prevents from accidental editing as it blocks the selection of objects of this type,
- Desired Speed Decisio the *insert mode* from which you can create and edit objects,

- A the *label mode*, which shows the label as defined in *graphic parameters*,
- and the *graphic parameters*, which specify visual settings on how objects are displayed in the *network editor*.

In the *quick view* window, the attributes of a specific network object can be displayed and edited. *Lists* are used to display and edit all the items within the same object type, even for non-graphic network objects such as *vehicle types* and *driving behaviors*.

4 Vehicle inputs

Vehicle inputs are associated with the demand in a Vissim road network, namely the number of vehicles that will travel through the network. The vehicle inputs are located at the start of each link that enters the network, and these are expressed in vehicles per hour. Toggle the *visibility* of the *Vehicle Inputs* object type. You can see now that every link entering the network is assigned a vehicle input (indicated by a black bar at the link start).

Next is to define our traffic demand, that is, how many vehicles and pedestrians are expected to interact at this intersection. These have already been defined and can be accessed by going to the *Network Objects* sidebar and then right-clicking on *Vehicle Inputs - show list*. To change or verify the vehicle inputs click on a row in the vehicle input list or double click on a specific vehicle input, e.g. the one at location G with id 11. This will open the *Vehicle Inputs* lists at the *list* window. Verify the total *Volume* entering at the west junction leg (location G), according to Figure 2.

To specify the composition of the traffic flow, go to *Traffic-Vehicle Composition*. Here, you can see two different vehicle compositions: the *Default* consisted of 95% *Car* and 5% *HGV* and the *Ped* with 100% pedestrians. You should select the *Default* vehicle composition for the vehicle links. Now, you should also add inputs for the pedestrian links (links 17 and 18). Set the volume at around 25 and select the appropriate vehicle composition.

The table now shows varying incoming traffic in intervals of 900 seconds each (15 min). Take a look at the table, there are 14 inputs, 4 for vehicles coming from each direction, and 8 for pedestrians (2 directions for each of the 4 crossings) ansd 2 for bike paths along Kunsgatan.

The routes for vehicles are also needed, that is, how many vehicles go straight ahead or turn. These have also been defined and can be accessed by going to the *Network Objects* sidebar, right-click on *Vehicle Routes - Show List - Static*. Take a look at the table, it now shows the routes for each of the 4 flows of vehicles, and the fraction of vehicles turning or going straight ahead.

5 Signal Control

Vissim offers a generic interface for signal controllers of various types. In this exercise, you will use a simple fixed-time controller,

First, we need to identify the possible conflicts in the intersection to assign green and red stages correctly. The following stages are defined (fig. 4).

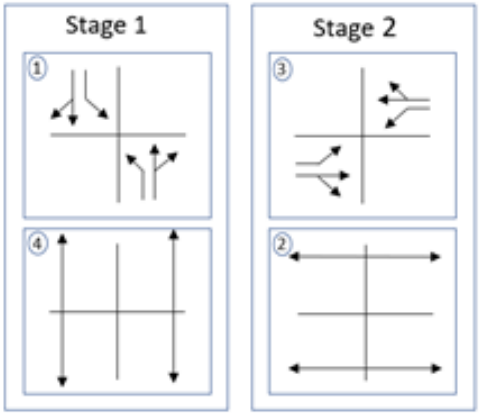


Figure 4: Signal groups at each stage.

When stage 1 has green light, stage 2 has red, and vice-versa. The top groups (1,3) in each stage refer to lights for vehicles, while the bottom groups (4,2) refer to lights for pedestrians.

An existing conflict in this design is the interaction between vehicles turning and pedestrians. In our model, this conflict is solved by assigning a preference to pedestrians. Another solution could have been to define an additional stage in which all vehicles are given red light while green to all pedestrians.

Now we can set up our signal controllers. In this exercise, we will compare a fixed-time controller against a vehicle-actuated controller.

5.1 Setting up a fixed-time signal controller

The term fixed-time signal controller is used when the green times of each stage, as well as the cycle time, is constant.

Click on *Signal Control – Signal Controllers* to display the list of the signal controllers. Right-click on the only row of the signal controller and select *Edit*. A dialog box will open, which contains the specifications of the signal controller to be implemented. To implement a fixed-time controller, select the type of signal controller *Fixed Time*. A fixed-time controller requires a VISSIG supply file, which you can find in the given files for this exercise (*"Fixed_time.sig"*).

Click on *Edit controller data* so the VISSIG window opens. To load the (*"Fixed_time.sig"*) click *File - Import*. Expand the *My signal control 1* object tree and then expand the *Signal programs* tree, and select Program No. 1 called *Kungsgatan - Stage 1, Slottsgatan - Stage 2*. Here (fig. 4) you can see the sequence of the traffic lights and the total cycle time of 90 seconds. Vehicles have a sequence of Red-red/amber-green-amber, while pedestrians have a sequence of Red-green. The shaded areas represent the transition from one stage to the other.

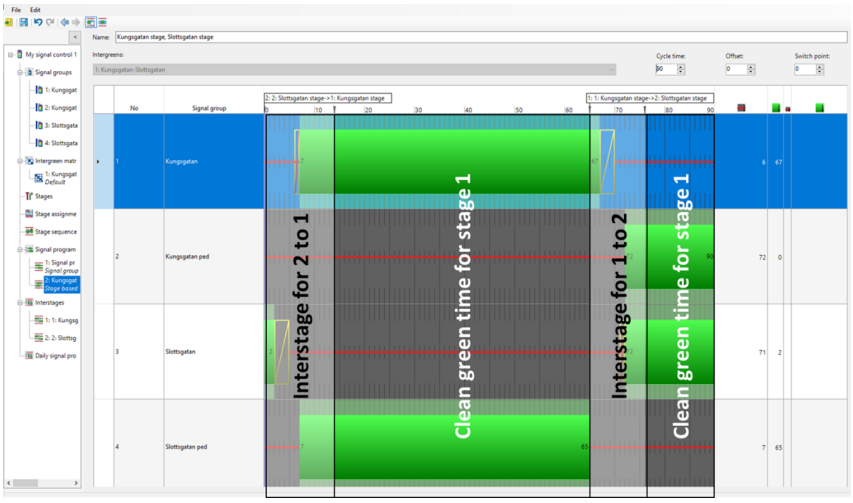


Figure 5: Signal groups at each stage.

You can now close this window by clicking on the *Back to Vissim* icon ( ) on the top left, then click *OK* on the *Signal Controller* window, and we are ready to test and run a simulation.

To run the simulation go to the top menu and click on *Simulation - Continuous*. Alternatively, click on the blue play icon ( ). The simulation will now start and can be visualized on the *Network Editor*. The simulation can be accelerated by moving the slider ( ) or by setting the simulation to maximum speed by clicking on the *Quick Mode* icon ( ). You might get a warning message at the end of the simulation, read what happened and dismiss it by clicking on *Ignore*. When the simulation ends it is set so that the metric of *average vehicle delay* is shown at the center of the intersection. The average vehicle delay (in seconds) is the average delay each vehicle will experience at the intersection. Remember this result.

5.2 Setting up an actuated signal controller

While the design of a fixed-time traffic signal controller is done offline and based on historical data, green times relying on an actuated signal controller are adaptive and respond to the current traffic conditions. In an actuated traffic signal control, the cycle time is not fixed, and the sequence of the stages can also be changed. Actuated traffic controllers respond to the presence of vehicles or pedestrians at the intersection, and might be better suited to control intersections where traffic volumes vary greatly. The presence of vehicles and pedestrians can be identified by detectors or even manually activated.

If you turn on the visibility of the *Detectors* object type from the *Network Object* sidebar, the location of each detector is displayed on the *Network editor*, and you should be able to see 8 detectors represented by small square boxes drawn on the road surface (see fig. 6)



Figure 6: Detectors.

A simplified explanation of the controller logic is as follows: once a vehicle is detected in either of these detectors, a signal is sent to the controller requesting either an extension of green light or a transition to a green light depending on the current state of the light. Under the hood, the logic includes constraints and conditions that need to be met to actually provide a green light. Defining this logic is out of the scope of this exercise, it requires the implementation of an algorithm using a scripting tool called VisVAP. For this exercise, the logic and required files are already provided and need to be added to our model.

Click on *Signal Control – Signal Controllers* to display the list of the signal controllers. Change the *Type* from *Fixed time* to *VAP*. Name the controller to 'Actuated controller' As *Logic File* select the file *Control_logic.vap*, then as *interstage file* select the file *Signal_data.pua* and click *OK* to close the *Signal Controller* dialog box.

Navigate to *Evaluation – Window – Signal Times Table* and select the *2: Actuated cntroller* as the selected signal controller. You are now ready to test and run the actuated traffic signal control.

6 Generating and evaluating results

Depending on the network objects used, various data is produced during a Vissim simulation, e.g., information on vehicles, links, areas, nodes, traffic jams, green time distribution, etc. Before running the simulation, you should first define and select the desired outputs according to the simulation's purpose. Then, the selected simulation outputs are displayed in evaluation lists.

In this exercise, we will use three evaluation objects: *Vehicle network performance*, *Data collections*, and *Queue counters*. The object *Vehicle network performance* contains information for specific attributes of the entire network, such as average vehicle speed or average delay. A delay measurement is derived by subtracting the travel time of a trip without any vehicle interaction, signal controls, or other required stops from the corresponding observed simulated trip travel time. You can also obtain results for specific network locations from the individual *Data collections Points*. *Data collection points* can be similar to the stationary detectors, e.g., inductive loop detectors, collecting cross-sectional data such as time-average flow and speed at

specific locations. Additionally, *Queue counters* can also be placed at specific locations. Vissim calculates some queue attributes, such as queue length, from the upstream position of a queue counter up to the last vehicle that has entered the queuing conditions.

Toggle the *visibility* of the *Data Collection Points* and *Queue Counters* object types. You can now see a data collection point at the east junction leg (orange bar), named *Slottsgatan East*, and a queue counter at the south junction leg named *Kungsgatan South*. From the *Evaluation* menu, select *Measurement Definition*, and then *Data Collection Measurements*. A list window opens, displaying an empty list. Add a row in that list by clicking on the green cross, and select the *Slottsgatan East* collection point under the column *DataCollectionPoints*. Next, from the *Evaluation* menu, select *Configuration*, and then select the *Results Attributes* tab. Check the *Vehicle network performance*, *Data collections*, and *Queue counters* at the *Collect data* column, and set the interval to 1800. Confirm with *OK*. In this way, you will be able to show the results aggregated for every 300-second (5-minute) time interval.

7 Run a simulation!

You are now ready to run a simulation by clicking the blue play button ( ). You can also switch from 2D to 3D view by clicking on the icon . By clicking on *Simulation - Parameters*, you can set parameters of the simulation such as the simulation period, starting time, and resolution.

- Change the simulation resolution from 10 to 20 and notice the difference.
- Now that you have run a simulation, what do you think? Based on the animated simulation, can you verify if the signal controller and the conflict areas are correctly added to the model?

If you need a quick simulation run without animation, you can use the *Quick Mode* by clicking on the icon before running the simulation. After a complete simulation, ignore any warning and navigate to *Results Lists* from the *Evaluation* menu. Select *Network Performance (Vehicle) Results*, *Data collection Results* and *Queue Results*. A list showing the results for each corresponding evaluation object will be displayed in the list window.

8 Calibrating the model

As in all simulation projects, the calibration of the model is very important. To calibrate the model, you have access to measurements of mean speeds and accelerations from 10 weekday mornings (see *traffic_measurements.xlsx*). **Important:** The data has been collected for the current signal control design; thus, the measurements correspond to the non-optimized fixed-time stage-based signal controller with fixed traffic demand.

To calibrate the model, switch the signal controller to the current situation, i.e., the fixed-time signal controller and program no. 1 (1: Fixed time controller). **Important:** it may be that you need to import the *Fixed_time.sig* file again; you can do so in the VISSIG window and then *File / Import*.

Measurements on speed and acceleration have been collected at 12 locations, which correspond to the data collection points in Vissim. Calibrate your model by repeatedly performing the following steps:

1. Run a suitable number of replications of the simulation.
2. Compare the simulation results and the measured data. This will help you define what parameters you might need to change.
3. Adjust one model parameter at-a-time, run simulation, and compare with measured data. Some examples of parameters you could modify include:
   - Average standstill distance
   - Additive factor of safety distance
   - Multiplicative factor of safety distance
   - Desired speed distribution
   - Desired acceleration function
   - Desired deceleration function

   To adjust the parameters related to car-following, you can navigate through them under *Base Data – Driving Behavior*, at the *Urban (motorized) behavior* settings. Moreover, you can find/modify the speed distributions under *Base Data – Distributions / Desired Speed*; acceleration functions are also available under *Base Data - Distributions*. You can also specify the different speed distributions for each type of vehicle through *Traffic – Vehicle Compositions*. Before changing a parameter, always think about the change you expect in the simulation.
4. Repeat the steps until you are satisfied with the comparison of the simulation results and the measured speeds.

Note that the Excel document provides an example where 5 replications were arbitrarily chosen to obtain the simulation speed for the collection point *Slottsgatan_East_in*. Therefore, you should not use these values, and instead, replace them with your simulation results. Then, you can compare the simulation results and the measured data.

To discuss: Before starting your calibration process, think about what model you are calibrating, what parameters you are changing, and what changes do you expect when changing a specific parameter.

To run multiple replications, navigate to *Simulation – Parameters* and change the *Number of runs* in the dialog box. Make sure you set the *Random seed increment* to a random number. This increment is the difference between two random seeds values when you perform multiple simulation runs. The value of a random seed initializes a random number generator. Two simulation runs using the same network file and seed generate the same results. If you vary the random seed, the stochastic functions in Vissim are assigned a different random value sequence leading to different simulation results. A comparison of these simulation results allows you to compare the effect of stochastic variations. For this purpose, Vissim calculates additional,

meaningful values for various result attributes during its evaluations, e.g., minimum value, maximum value, and mean over all replications.

To discuss: How many replications do you need to run for your calibration?

9 Simulating with Scenario Management

When using traffic simulation to assess changes in a traffic system, one would like to test various scenarios reflecting different traffic conditions. Typically, you start with a *baseline scenario* representing the current state of the network; this would be your calibrated modelof the current design with a fixed time signal controller. Additional scenarios are then created to analyze the impact of specific modifications, such as changes in the vehicle composition, network geometry, or–in this exercise– the implementation of a new traffic signal controller at the intersection or changes in traffic demand.

Instead of changing traffic conditions manually for each new scenario, Vissim's Scenario Management functionality allows you to efficiently set up, run, and compare multiple scenarios within a single project. This approach facilitates the simulation process and provides an organized comparison of results.

1. Make sure you have the signal controller to the current situation, i.e., the fixed-time signal controller and select the program no. 1 (Fixed time).

2. Before using Vissim's Scenario Management, ensure you have your desired simulation parameters (simulation duration, random seed, and number of runs), and create a copy (all files) of the your calibrated model as a backup.

3. Open the *Scenario Manager* from the main menu, under *File*. Then, select *Place under scenario management*. The new window will ask you to confirm the creation of the first scenario. Name the first scenario as "baseline".

4. A new window will appear, which shows the new structure of the project. In this structure, you will see the *Base network* and a folder named *Scenarios*. The *Base network* correspond to the core geometry and attributes that are common to all scenarios. In principle, you should not modify this network, unless you need to modify a model feature that applies to all scenarios. In other words, any changes made to the *Base network* will affect all scenarios. **Important:** The *Base network* itself is not a scenario and will not be simulated directly.

5. Right-click on the folder *Scenarios*, and select *Edit project structure*. You will now see a window with three tabs: 1) *Basic settings* which keeps a record of all changes you make, 2) *Scenarios* showing a list of all created scenarios, and 3) *Modifications* where you define all modifications you are interested in simulating. A modification is a set of traffic conditions, and a scenario can have one or more modifications.

    - **Important:** Before you start creating new scenarios, decide how many scenarios you will investigate.

6. Once you decide which scenarios to simulate, start by creating the modifications under the *Modifications* tab. For example:

    - One modification could be to switch the signal controller to the actuated signal controller. Write a suitable name, description and group for this modification. Another modification could be changing the traffic demand. Add as many modifications as you need. For this exercise, do not add any exclusions or dependencies.

7. Once you create the modifications, go to the *Scenarios* tab. Here you will add as many scenarios as you want, but keep in mind that the first (already created) scenario corresponds to the baseline, so we recommend keeping this scenario without any modifications. Feel free to write a suitable name and description for each scenario.

8. For each scenario, define which modifications are going to be implemented by selecting them from the column titled *Modifications*. The structure of multiple-scenario simulation is now set, close the window.

9. Now, it is time to define the traffic conditions for each modification. In the *Project Explorer* window, you now see the modifications you added under the folder *Modifications*. To make the corresponding changes in the traffic system, double-click on the first "modification" listed in that folder, and it will open an identical version of your *Base network* model. In the *Modification 1* version, change the traffic system according to the purpose of this "modification". For example:

    - To change the signal controller, open *Signal Control - Signal Controllers*, and adjust as needed.
    - To add variable traffic demand in time intervals, open the *Vehicle Inputs* list and fill in different traffic flow values for each time interval; uncheck the *Continued* option.
    - **Important:** Edit the *Modification* directly, rather than making edits within scenarios.

10. Once you finish defining all modifications, you are now ready to run all scenarios. Right-click on the folder *Scenarios* and select *Simulate all scenarios*. Alternatively, you can also run one scenario at-a-time, but you need to make sure to first open the scenario you want to simulate.

11. By default, the simulation results are stored in every scenario. For personalized analysis, you can copy and paste results directly from the *Results lists*. Alternatively, you can directly compare one scenario against other in Vissim. For example:

    - Go to *Edit project structure*, and you will see a column titled *ScenToComp*. In this column, select which scenarios you wish to compare with, e.g., the baseline could be compared with all modified scenarios.
    - After running the simulation, go to the *Data Collection Results list* and add a new attribute. In the attribute list, find *Scenario comparison Scenario 1 / Scenario 2 - Scenario 1 % / Queue delay / All types*, and add by clicking on *Add from all scenario comparisons*, and confirm. Look at a new columns in the *Data Collection Results list*.

10 Comparing different alternatives

Choose and conduct one of the following alternative design analaysis

1. Evaluate the impact of the different signal traffic controller on the intersection's traffic performance. For this, you will simulate traffic at the intersection under the following scenarios:
   - Current situation (base scenario), i.e., a fixed-time stage-based signal controller.
   - An actuated traffic signal controller.
2. How would an incident/accident potentially affect the traffic performance at the intersection? To answer this question, you will consider two demand scenarios:
   - Constant flow: The input vehicle rates are constant for the entire simulation period.
   - Variable flow: We assume that due to a possible incident, the vehicle input rates change significantly every 10 minutes according to Table 1.

Table 1: Vehicle flow demand [veh/h] per time interval.

Time interval (minutes)	A	C	E	G
0-10	670	160	1100	100
10-20	640	370	920	380
20-30	670	190	1100	150

3. How would an increase in bicycle traffic demand affect the traffic performance at the intersection? To answer this question, consider a future scenario in which bicycle traffic volumes increase to 800 bicyclists (in each travel direction). Assume a constant flow for the entire simulated time period.

11 Report

Before next seminar you should send a short report on your work (deadline 12th of October). The report should contain a

- a description of your calibration process (what parameters did you change and why), how many replications you ran, and what was the outcome of the calibration. HINT: To present the numerical results of the calibration, you can include figures or tables with confidence intervals to illustrate the comparison between the collected measurements and your calibrated model, in all collection points. In the provided Excel file, you will find a template for comparing data as a guide, but feel free to modify it according to your needs or use other tools. Keep in mind to always annotate the unit of the chosen metric when presenting the results.
- a description of which alternative designs you decided to study,
- simulation results (figures and/or tables) and comparison between the alternatives. Keep in mind that the choice of traffic metrics for this analysis should be motivated and described.

- discussion on the result
- reflections of the exercise in relation to the previous seminars.

Also prepare and send (same deadline as for the report) a couple of slides including

- 1-2 slides on what parameters you changed and why and what was the outcome of the calibration
- 1 slide with a short description of which alternative designs you decided to study
- 2 slides with the most interesting simulation results (figures and/or tables)
- 1 slide with observations and/or questions that you want to discuss